

Probing the Structural Dependence of Carbon Space Lengths of Poly(*N*-hydroxyalkyl acrylamide)-Based Brushes on Antifouling Performance

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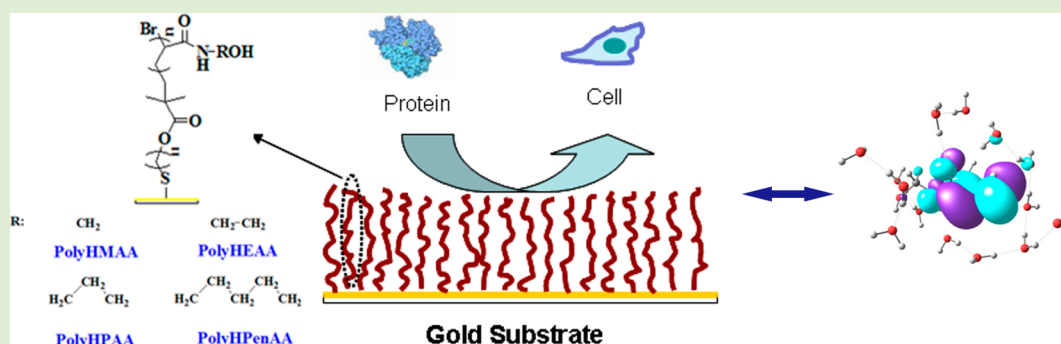
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Supporting Information



ABSTRACT: Numerous biocompatible antifouling polymers have been developed for a wide variety of fundamental and practical applications in drug delivery, biosensors, marine coatings, and many other areas. Several antifouling mechanisms have been proposed, but the exact relationship among molecular structure, surface hydration property, and antifouling performance of antifouling polymers still remains elusive. Here this work strives to provide a better understanding of the structure–property relationship of poly(*N*-hydroxyalkyl acrylamide)-based materials. We have designed, synthesized, and characterized a series of polyHAAA brushes of various carbon spacer lengths (CSLs), that is, poly(*N*-hydroxymethyl acrylamide) (polyHMAA), poly(*N*-(2-hydroxyethyl)acrylamide) (polyHEAA), poly(*N*-(3-hydroxypropyl)acrylamide) (polyHPAA), and poly(*N*-(5-hydroxypentyl)acrylamide) (polyHPenAA), to study the structural dependence of CSLs on their antifouling performance. HMAA, HEAA, HPAA, and HPenAA monomers contained one, two, three, and five methylene groups between hydroxyl and amide groups, while the other groups in polymer backbones were the same as each other. The relation of such small structural differences of polymer brushes to their surface hydration and antifouling performance was studied by combined experimental and computational methods including surface plasmon resonance sensors, sum frequency generation (SFG) vibrational spectroscopy, cell adhesion assay, and molecular simulations. Antifouling results showed that all polyHAAA-based brushes were highly surface resistant to protein adsorption from single protein solutions, undiluted blood serum and plasma, as well as cell adhesion up to 7 days. In particular, polyHMAA and polyHEAA with the shorter CSLs exhibited higher surface hydration and better antifouling ability than polyHPMA and polyHPenAA. SFG and molecular simulations further revealed that the variation of CSLs changed the ratio of hydrophobicity/hydrophilicity of polymers, resulting in different hydration characteristics. Among them, polyHMAA and polyHEAA with the shorter CSLs showed the highest potency for surface hydration and antifouling abilities, while polyHPenAA showed the lowest potency. The combination of both hydroxyl and amide groups in the same polymer chain provides a promising structural motif for the design of new effective antifouling materials.

1. INTRODUCTION

Bioinert or antifouling polymers have been widely used for many scientific and technological applications including blood-

Received: April 23, 2014

Revised: June 23, 2014

Published: June 26, 2014

contacting biomaterials, biosensors, drug delivery carriers, and even ship hulls.^{1–3} Significant efforts and progress have been made to design and synthesize a wide range of bioinert polymers including poly(ethylene glycol) (PEG),⁴ poly(hydroxyethyl methacrylate),⁵ poly(acrylamide),⁶ polysaccharides,⁷ and polypeptoids.⁸ These bioinert/antifouling polymers vary significantly in molecular structures, charge distributions, and chemical groups, but they all achieve a similar function of preventing nonspecific interactions between polymers and biomolecules.⁹ Such nonspecific polymer–biomolecule interactions often cause adverse immune responses for implanted devices and drug delivery vehicles,¹⁰ compromise detection sensitivity and efficacy for biosensors,¹¹ and increase energy and environmental costs for marine coatings.¹² Among bioinert and antifouling polymers, the most successful ones are PEG-based materials. However, PEG-based materials are susceptible to oxidation in most of biochemical media containing transition metal ions, limiting their applications.¹³ Recently, zwitterionic-based materials, such as poly(2-methacryloyloxyethyl phosphorylcholine) (polyMCP),¹⁴ poly(sulfobetaine methacrylate) (polySBMA),¹⁵ and poly(carboxybetaine methacrylate) (polyCBMA),¹⁶ have demonstrated their excellent bioinert capacity and high stability in many complex media including undiluted human blood serum/plasma.

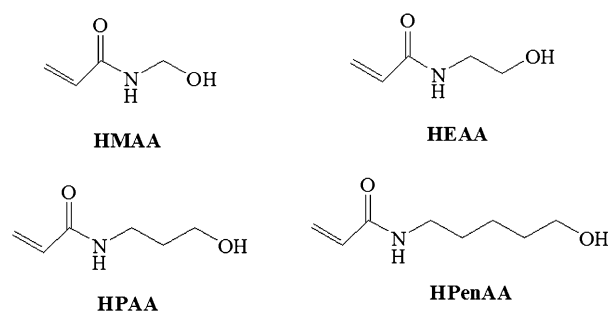
Several bioinert or antifouling mechanisms have been proposed. Upon coating of antifouling polymers on the surface, a nonfreezable water layer is formed at polymer interface, which provides a physical and energetic barrier to prevent biomolecules from in contact with the surface. Thus, surface hydration is often considered to be a primary factor for controlling antifouling performance. Because all existing antifouling polymers are either hydrophilic or zwitterionic, the hydration layer at different polymer interfaces can be achieved differently; that is, hydrophilic polymers achieve surface hydration via hydrogen bonds, while zwitterionic polymers achieve hydration via ionic solvation.^{17,18} Moreover, the surface coated with flexible and elastic polymers can also sterically repel approaching biomolecules.¹⁹ Generally speaking, intrinsic chemical properties of polymer itself (i.e., molecular structure, chemical groups, and chain length) and interfacial architecture (packing density, film roughness, and film thickness) of polymer coating are equally important for antifouling surfaces. Despite different antifouling mechanisms, it is not likely that a single mechanism is solely responsible for the onset of antifouling events but rather a combination of many. In particular, among other antifouling factors, a molecular level understanding of the dependence of polymer structure on antifouling performance is a hardcore problem, which is extremely important for fundamental understanding of antifouling mechanisms and practical development of next-generation antifouling materials.

We have developed systematic methodologies to synthesize and prepare a series of zwitterionic and hydrophilic antifouling polymers in different forms of polymer brushes, hydrogels, nanogels, and nanoparticles.^{20–25} All of them exhibited strong surface resistance to proteins, cells, and bacteria. Until very recently, we started to investigate the structural dependence of polymers on antifouling performance.²⁶ We synthesized two different types of polyacrylamide polymers (polySBAA and polyHEAA) and two corresponding polyacrylate polymers (polySBMA and polyHEA). Two polyacrylamide monomers only differ from the corresponding polyacrylate monomers by substitution of O group with NH group in the backbone.

Surface plasmon resonance (SPR) data have showed that although all four polymer brushes were highly resistant to protein adsorption from undiluted human blood serum/plasma, the polySBAA and polyHEAA with amide groups (<0.3 ng/cm²) exhibited better protein resistance than the polySBMA and polyHEA with oxygen groups (3.5 – 7.5 ng/cm²). This confirms that a subtle structural change in polymer backbones/side chains allows us to greatly improve antifouling performance of biomaterials.

Motivated by our previous findings, here we synthesized and characterized four poly(*N*-hydroxyalkyl acrylamide)-based polymers with different carbon spacer lengths (CSLs): poly(*N*-hydroxymethyl acrylamide) (polyHMAA), poly(*N*-(2-hydroxyethyl)acrylamide) (polyHEAA), poly(*N*-(3-hydroxypropyl)acrylamide) (polyHPAA), and poly(*N*-(5-hydroxypentyl)acrylamide) (polyHPenAA). HMAA, HEAA, HPAA, and HPenAA monomers contained one, two, three, and five methylene spacer groups between hydroxyl and amide groups, respectively, while the other groups in polymer backbone are identical to each other (Scheme 1). The presence

Scheme 1. Chemical Structure of HMAA, HEAA, HPAA, and HPenAA Monomers



of both amide group and hydroxyl group in one molecule tends to improve surface hydration ability of polymer brushes; meanwhile, different CSLs between amide group and hydroxyl group are expected to affect hydrogen bond formation between water and poly(*N*-hydroxyalkyl acrylamide) molecules due to specific geometrical criteria for hydrogen-bond formation, suggesting that there may be existence of an optimal CSL for achieving the best surface hydration. Moreover, different CSLs would also change the hydrophilicity/hydrophobicity ratio of the polymers, resulting in different water–polymer interactions that would induce polymer conformation change and hydrogen bonding formation. Different CSLs can also affect the flexibility of polymers that will affect their water binding ability. The CSL-induced combination effects would result in different surface hydration and antifouling behaviors for different polymer brushes. Herein, we combined different computational and experimental approaches to examine the effect of CSLs as a structural factor on the hydration and antifouling ability of poly(*N*-hydroxyalkyl acrylamide)-based polymers. SPR and cell assay were used to real-time monitor surface resistance to proteins and cells on polymer brushes, while sum frequency generation (SFG) vibrational spectroscopy and molecular dynamics simulations were used to quantify the structure and dynamics of water at polymer interface. The effect of film thickness on antifouling performance was also examined. This work provides a better fundamental understanding of a structure–property correlation among molecular structure, surface hydration characteristic, and antifouling performance

of the polymers, particularly CSL effect, which hopefully helps to rationally design new effective antifouling polymers.

2. MATERIALS AND METHODS

2.1. Materials. *N*-Hydoxymethyl acrylamide (HMAA, 48% solution in water), *N*-2-hydroxyethyl acrylamide (HEAA, 97%), 3-amino-1-propanol (99%), 5-amino-1-pentanol(95%), acryloyl chloride (97%), copper(I) bromide (CuBr), and phosphate buffer saline (PBS, pH 7.4, 0.15M, 138 mM NaCl, 2.7 mM KCl), were purchased from Sigma-Aldrich (Milwaukee, WI) and used as received. Tris[2-(dimethylamino)ethyl]amine (Me6TREN, 99%) was purchased from Alfa Aesar (Ward Hill, MA). ATRP initiator for gold surface, ω -mercaptoundecyl bromoisobutyrate, was synthesized through the reaction of bromoisobutyl bromide and 11-mercapto-1-undecanol using a method published previously.²⁷ The initiator for silica prism, 3-(2-bromoisobutyramido) propyl(trimethoxy)silane, was purchased from Gelest (Morrisville, PA). Water used in these experiments was purified by a Millipore water purification system with a maximum resistivity of 18.0 M Ω cm. All other reagents and solvents were commercially obtained at extra-pure grade and were used as received. SPR glass chips were coated with an adhesion-promoting chromium layer (2 nm in thickness) and a plasmon active gold layer (48 nm) by electron beam evaporation under vacuum.

2.2. Synthesis of HPAA and HPenAA. Two monomers, HPAA and HPenAA, were synthesized and purified using the method published previously.²⁸ In brief, 75 mL of anhydrous ethanol was added to a 250 mL three-necked flask equipped with a thermometer, a funnel drop, and a nitrogen inlet and cooled to -40°C . 0.05 mol of acryloyl chloride was also cooled to -40°C and then slowly added to the flask. Subsequently, solution of 0.1 mol of 3-amino-1-propanol or 5-amino-1-pentanol in 75 mL of ethanol was added in a dropwise way for ~ 1 h, during which temperature was maintained between -45 and -40°C . After feeding, the reaction mixture was stirred for 2 h at -40°C , followed by additional stirring for 12 h at 5°C . As the solvent was evaporated, the residue was dissolved in acetone. After filtering the precipitates, the resultant solution was purified by a silica gel column with acetone as an eluent. The eluent was stabilized by adding 20 ppm hydroquinone and then evaporated at 0°C under vacuum. The products were collected and dissolved in Millipore water forming 25 wt % solution and stored at 4°C . The pure products were analyzed by NMR. ^1H NMR 300 MHz (D_2O , δ): HPAA: 6.13 (dd, 1H, =CH), 6.09(d, 1H, CH_2 =), 5.57 (d, 1H, CH_2 =), 3.55 (t, 2H, CH_2), 3.25(t, 2H, CH_2), 1.70(m, 2H, CH_2). HPenAA: 6.11 (dd, 1H, =CH), 6.09(d, 1H, CH_2 =), 5.57 (d, 1H, CH_2 =), 3.50 (t, 2H, CH_2), 3.18(t, 2H, CH_2), 1.47 (m, 4H, CH_2), 1.27(m, 2H, CH_2).

2.3. Preparation of Polymer Brushes via the SI-ATRP Method. Au-coated SPR chips were cleaned by ultrasonication sequentially in acetone, isopropanol, and water and then exposed to UV light in ozone atmosphere for 15 min to further remove organic impurities absorbed on the chips. Subsequently, the chips were rinsed by ethanol and water and dried by nitrogen steam. In Scheme 2, to anchor initiators on SAM surface, freshly cleaned chips were placed

into 1.0 mM ethanol solution of ω -mercaptoundecyl bromoisobutyrate at room temperature for 24 h. Then, in the first reaction tube, 15.7 mg of CuBr (0.11 mmol) and SPR chip with initiator-coated SAM surface were placed into a tube under evacuation and filling with nitrogen three times to remove oxygen. After deoxygenating, 5.0 mL of degassed methanol was added. In the second reaction tube, monomer (0.5 g) and Me6TREN (40 mg, 0.174 mmol) were dissolved in Milli-Q water (5 mL), and the mixture was degassed with nitrogen for 15 min. The solution was then transferred to the first reaction tube using a syringe under nitrogen protection. The tube was then completely sealed for ATRP polymerization at room temperature. Different reaction times of ATRP polymerization allow us to control and achieve different film thicknesses of polymer brushes. After the reaction, the chips were removed and repeatedly rinsed by methanol, water, and buffer solution. The chips grafted with polymer brushes were soaked in buffer solution overnight before SPR measurement. Silica prisms were first cleaned by piranha solution, dried by nitrogen stream, then placed into 10 mM initiator solution with dehydrated toluene for 24 h to form initiator monolayer to graft polymer brushes on silica prism for SFG measurement. Upon the formation of initiator monolayer, the following ATRP procedure for polymer brushes on silica prisms was the same as the one for polymer brushes on SPR chips.

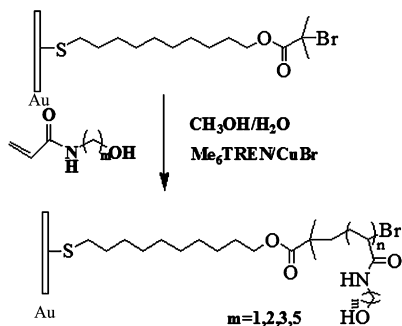
2.4. Polymer Thickness by Ellipsometry. The film thickness of polymer brushes was measured by an α -SE ellipsometer (J.A. Woollam, Lincoln, NE) with a He-Ne laser ($\lambda = 632.8$ nm) and a fixed angle of incidence of 70° . Bare SPR sensor chip was first used to determine the thickness and optical parameters of Ti layer and Au layer. The thickness and refractive index of polymer brushes was achieved by fitting the data using Cauchy model with predetermined parameters of bare SPR sensor chip.

2.5. Contact Angle. Water contact angles on different surfaces were recorded with Rame-Hartgoniometer (model 100-00, Mountain Lake, NJ), equipped with a sessile drop-shape analysis system and a video camera. A 5 μL droplet of water was placed on the surface by microsyringe for the static contact-angle measurement. The data presented were averaged by five independent measurements on different positions.

2.6. Nonspecific Protein Adsorption by SPR. Protein adsorption was measured by a custom-built SPR sensor, which is based on wavelength interrogation. A peristaltic pump (Ismatec) was utilized to deliver liquid sample to four independent channels of the flow cell. After being rinsed with PBS overnight, the SPR chip was dried with air and attached to the base of the prism. A baseline signal was established by flowing PBS buffer at a rate of 0.05 mL/min through the sensor until the baseline being stable. One mg/mL single protein solution of fibrinogen, BSA, lysozyme, and undiluted blood serum and undiluted blood plasma were flowed through independent channels for 10 min, followed by PBS buffer solution to remove unbound or loosely bound proteins. The amount of adsorbed proteins on the surface was quantified by measuring wavelength change between the baselines before and after protein injection. One nm SPR wavelength shift at 750 nm represents ~ 15 ng/cm 2 adsorbed proteins. Because we used a four-channel SPR sensor, three channels were used to measure protein adsorption from the same protein solution and one channel was used for the control experiment. Thus, each SPR data is sampled by three independent channels and by two independent experiments.

2.7. Hydration of Polymer Brushes by SFG. A commercial SFG system purchased from EKSPLA was used in this study, and the optical setup has been reported in detail previously.^{29–31} The output of the Nd:YAG laser is a 1064 nm near-IR beam (20 Hz, 20 ps). A frequency tunable mid-infrared (mid-IR, wavenumber range: 1300 to 4300 cm^{-1}) laser beam and a 532 nm visible laser beam were generated from this Nd:YAG laser. The visible laser beam and IR were overlapped at the sample interface spatially and temporally. The incident angles of the visible and the IR input beams were 60 and 55° with respect to the surface normal, respectively. The diameters of the two input beams at the surface were ~ 500 μm . The SFG spectra were collected in the SSP (s-polarized signal output, s-polarized visible input, and p-polarized IR

Scheme 2. Preparation of Poly(*N*-hydroxyalkyl acrylamide)-Based Brushes on a Gold Surface via SI-ATRP



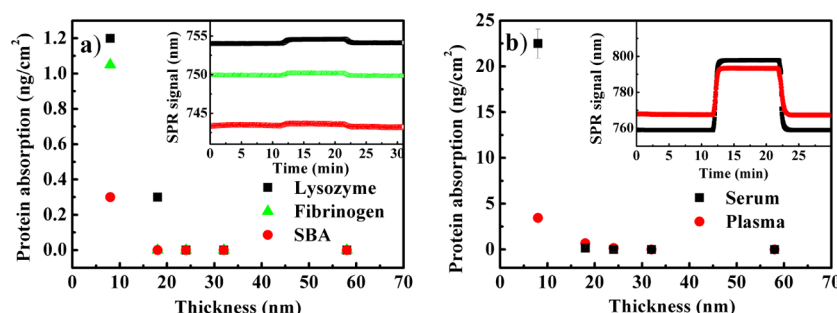


Figure 1. Protein adsorption of (a) single protein solutions of lysozyme, fibrinogen, and BSA (1.0 mg/mL) and (b) undiluted human blood plasma and serum on polyHMAA brushes as a function of film thickness. Inset: the original SPR sensorgrams for the polyHMAA brushes with a film thickness of 24 nm.

input) polarization combination. For SFG spectrum, the signal intensity can be defined as

$$I_{\text{SFG}} \propto |\chi_{\text{eff}}^{(2)}|^2 I_{\text{IR}} I_{\text{vis}} \quad (1)$$

where I_{IR} and I_{vis} are the input IR and visible beam intensities, respectively. $\chi_{\text{eff}}^{(2)}$ is the effective second-order nonlinear susceptibility of the interface, which can be defined as

$$\chi_{\text{eff}}^{(2)} = \chi_{\text{NR}}^{(2)} + \sum_q \frac{A_q}{\omega_{\text{IR}} - \omega_q + i\Gamma_q} \quad (2)$$

Here $\chi_{\text{NR}}^{(2)}$ arises from the nonresonant background contribution, and the resonant contribution can be modeled as a sum of Lorentzian peaks with signal strength or amplitude, A_q , frequency, ω_q , and line width, Γ_q . Therefore, the SFG spectra can be fitted using eq 2. In this paper, two-peak and three-peak models are used to fit the spectra from the samples contact with water and dried samples, respectively.

2.8. Cell Adhesion. Bovine aortic endothelial cells (BAECs) were cultured in culture medium containing DMEM, 10% fetal bovine serum (FBS), 1% sodium pyruvate, 1% nonessential amino acids, and 2% penicillin streptomycin for 72 h at 37 °C. The adherent cells were detached by treatment with trypsin/ethylenediaminetetraacetic acid (0.05%/0.53 mM), washed with PBS, and diluted in the culture medium to reach a final concentration of 10^5 cells/mL. The chips for cell adhesion measurements were first sterilized by 10 min of UV explosion, washed with sterile PBS solution for three times, and then placed in individual wells of a 24-well plate. Two mL of cell suspensions was added to each well and incubated with the samples for 7 days, with culture medium refreshed every 3 days. Phase-contrast images (10X) were finally acquired using an EVOS XL core inverted microscope (Advanced Microscopy Group, Bothell, WA). Three independent samples of a given polymer brush were used for cell adhesion experiments.

2.9. Molecular Simulations of Polymers in solution. Force-field parameters for monomeric HMAA, HEAA, HPAA, or HPenAA molecules were developed using the ParamChem tool (<https://www.paramchem.org/>), which is compatible with the general CHARMM force field. After geometry optimization at MP2/6-31G* level by Gaussian95, the partial charges were derived by fitting to the gas-phase electrostatic potential using the restrained electrostatic potential (RESP) method. To validate the force-field parameters of monomeric HMAA, HEAA, HPAA, and HPenAA, we conducted short 2 ns simulations in the explicit TIP3P water at 298 K. The bond lengths, bond angles, and torsion angles were well-maintained in the MD simulations as compared with quantum-mechanism structures. The equilibrium structures of HMAA, HEAA, HPAA, and HPenAA with their first solvent shells were extracted and subjected to the 20 000 steps minimization before the quantum calculations. Density function theory (DFT) with Lee–Yang–Parr gradient-corrected correlation function (B3LYP) method and 6-31+(d,p) basis was employed to obtain the single-point energies and frontier molecular orbital for each monomer–water system at both HOMO (highest occupied molecular

orbital energy) and LUMO (lowest occupied molecular orbital energy) states.

MD simulations of a 16-mer of polyHMAA, polyHEAA, polyHPAA, or polyHPenAA in a cubic TIP3P water box of $60 \times 60 \times 60 \text{ \AA}^3$ were performed using the NAMD program³² with CHARMM22 force fields.³³ Each simulation system was first subject to 5000 steps of steepest decent minimization with position constraints on polymers, followed by an additional 5000 steps of conjugate gradient minimization without any position constraint. After energy minimization, each system was gradually heated from 50 to 298 K in 200 ps and equilibrated at 298 K for 1 ns to adjust the system size and density under NPT ensemble with periodic boundary conditions in all three dimensions. Then, 20 ns production MD run was conducted to examine the mutual dynamics and interactions between polymers and water molecules. Short-range van der Waals (VDW) interactions were calculated by a switch function with a twin cutoff at 10 and 12 Å, while long-range electrostatic interactions were calculated by the particle-mesh Ewald method with a grid size of 1 Å and a real-space cutoff of 14 Å. The RATTLE algorithm was applied to constrain all covalent bonds involving hydrogen atoms so that a time step of 2 fs was used in velocity verlet integration. Each polymer system was independently run three times for validation with the different starting coordinates and velocities. MD trajectories were saved by every 2 ps for analysis.

3. RESULTS AND DISCUSSION

3.1. Film Thickness Dependence of Protein Adsorption on polyHMAA Brushes. All four poly(*N*-hydroxyalkyl acrylamide)-based brushes were prepared by SI-ATRP method, according to Scheme 2. The thickness of polymer brushes can be tuned by controlling polymerization time. A number of our and other studies have shown that the thickness of polymer brushes affects their antifouling performance.^{24,25,34} We first determined the optimal film thickness and experimental conditions of polyHMAA polymer brushes, which will serve as basic parameters to prepare other poly(*N*-hydroxyalkyl acrylamide)-based brushes with similar film thicknesses for examining CSL effect. (See subsection 3.2.) Figure 1 shows the film thickness dependence of protein adsorption from both single protein solutions and undiluted human blood serum and plasma on polyHMAA brushes. The film thicknesses of polyHMAA brushes were varied from 8 to 60 nm, as controlled by ATRP polymerization time. First, three single protein solutions of lysozyme (14 kDa, pI 12), fibrinogen (340kD, pI 5.5), and BSA (68kD, pI 4.8) were used separately to examine nonspecific protein adsorption on polyHMAA brushes of different thicknesses. As shown in a typical SPR sensorgram (an inset of Figure 1a), the adsorption amount of proteins can be determined by a wavelength shift between the baselines before and after protein solution injection, where 1 nm wavelength shift is equivalent to 15 ng/cm², as calibrated in our previous

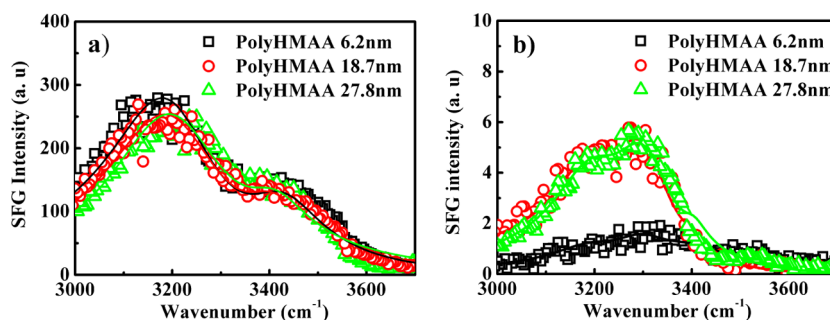


Figure 2. SFG spectra of polyHMAA brushes with film thicknesses of 6.2, 18.7, and 27.8 nm (a) at a wet state and (b) at a dry state. Solid curves are fitting to experimental data.

works.²⁴ The very thin polyHMAA brush of ~ 8 nm induced a small amount of the adsorption of lysozyme (1.2 ng/cm^2), fibrinogen (1.1 ng/cm^2), and SBA (0.3 ng/cm^2), respectively, likely owing to insufficient surface hydration and limited polymer elasticity. However, as film thicknesses were higher than 8 nm, SPR data showed that polyHMAA brushes exhibited almost undetectable nonspecific protein adsorption ($<0.3 \text{ ng/cm}^2$) from three single-protein solutions, which is considered to be an ultralow fouling surface (Figure 1a). We further challenged our polyHMAA brushes using undiluted blood plasma and serum. Similar trend of protein adsorption as a function of film thickness was observed. In Figure 1b, polyHMAA of ~ 8 nm thickness exhibited ~ 4.5 and 22.5 ng/cm^2 adsorbed proteins from undiluted blood plasma and serum, while polyHMAA of 18–60 nm thicknesses significantly reduced adsorbed proteins to a superlow fouling level of $<0.3 \text{ ng/cm}^2$. In all cases tested, polymer brushes need to arrive a threshold value of film thickness of >8 nm, which allows us to achieve large surface coverage, adequate hydration layer, and high steric repulsion and thus the better fouling capacity. While protein adsorption from single-protein solutions is not very sensitive to film thickness, protein adsorption from undiluted blood plasma and serum shows non-negligible dependence of film thickness. Similar film thickness effects were observed for other antifouling materials such as polySBMA,¹⁵ polyCBMA,¹⁶ poly(serine methacrylate),³⁵ and poly(hydroxy-functional methacrylates).²⁵

The state of water at both hydrated and dried polyHMAA-grafted surfaces of different film thicknesses (i.e., 6.2, 18.7, and 27.8 nm) was analyzed by SFG with SSP polarization (Figure 2). In SFG spectrum, signals in $3000\text{--}3700 \text{ cm}^{-1}$ region typically represent O–H stretching signals.³⁶ For the hydrated polyHMAA brushes of different film thicknesses, two characteristic peaks of interfacial water at 3200 and 3400 cm^{-1} were observed, indicating that water molecules are ordered at polyHMAA interfaces. We believe that the intensity of the peaks reflects the number of water molecules at interface; that is, the higher of signal intensity, the more surface-bound water molecules. Three hydrated polyHMAA surfaces displayed almost identical SFG profiles with similar peak locations and intensities. The almost independent of SFG intensity change on film thickness indicates that SFG intensity mainly arises from interfacial water on the brushes, not from water inside the brushes. Additionally, similar SFG profiles can not be directly used to quantify the extent of surface hydrations of different polyHMAA brushes at a wet state. Thus, to better associate SFG signals of interfacial water with surface hydration character, we designed dry SFG measurement on polyHMAA brushes. To obtain dry samples, we gently removed the

interfacial water on wet polyHMAA-grafted brushes at the bottom of prism by flowing nitrogen stream at the same flow rate (controlled by the open of valve) for 3 s, followed by immediate SFG measurement. In the case of all dry polyHMAA brushes (Figure 2b), the intensities of SFG signals were significantly reduced by at least an order of magnitude, in which the loss of SFG signals is well-correlated with the loss of interfacial ordered water upon drying. More interestingly, upon drying of polymer brushes, a clear dependence of SFG signals on film thickness was observed; that is, the amount of interfacial water showed increment trend with film thickness. Interfacial ordered water no longer existed for the thin polyHMAA brush of 6.2 nm, as evidenced by almost negligible SFG signals. However, a combination of three peaks centered at 3200 , 3300 (N–H signal from the polymer), and 3400 cm^{-1} was observed for the thicker polyHMAA brushes of 18.7 and 27.8 nm, indicating the existence of interfacial water. This indicates that the thicker polymer brushes not only effectively attract water molecules to their surface and inside but also protect interfacial water molecules, particularly those inside the brushes, from the drying process. The SFG analysis on the “dry” polymer brushes suggests that as compared with thin brushes the thicker brushes have the stronger surface hydration.

3.2. Effect of Carbon Spacer Length on Protein Adsorption. From a structural point of view, while the change of CSLs from methylene to pentyldene is expected to affect surface hydration and antifouling properties of polymer brushes, the exact role of CSLs in the structure–property relationship of antifouling polymers still remains elusive. Herein, to probe CSL effects on surface hydration and antifouling performance of poly(*N*-hydroxyalkyl acrylamide)-based polymers, we synthesized and grafted polyHMAA, polyHEAA, polyHPAA, and polyHPenAA onto the Au surface using the SI-ATRP method. Each of the four poly(*N*-hydroxyalkyl acrylamide)s has a distinct spacer group between hydroxyl and amide groups, that is, a methylene, an ethylene, a propylidene, and a pentyldene group. Film thickness and surface hydrophobicity of the prepared brushes were characterized using ellipsometry and water contact angles, respectively. The polymerization conditions (e.g., reaction time, catalyst, and solvent) were carefully selected and controlled to achieve similar film thickness of ~ 33 nm for four polymer brushes, as measured by ellipsometer (Figure 3). Our previous data suggest that this range of film thicknesses is likely to achieve minimum protein absorption for polyHEAA, polyHPMA, and polyHEMA brushes.^{20,24,25} Consistent with film thickness, the film refractive indexes of four polymer brushes were ~ 1.5 , as determined by fitting ellipsometer data (Figure 3). These data confirm that all brushes have comparable high

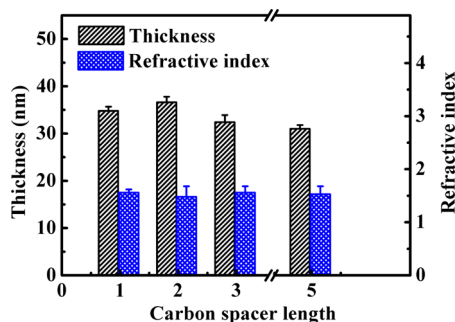


Figure 3. Average film thickness and refractive index of poly(*N*-hydroxyalkyl acrylamide) brushes of various carbon spacer lengths ($n = 3$).

packing density,³⁷ consistent with results of graft densities of the brushes obtained by the method of adding free initiator (Figure S1 and Table S1 in the Supporting Information)

To quantify surface wettability of polymer brushes, we show in Figure 4 water contact angles on the surfaces grafted with

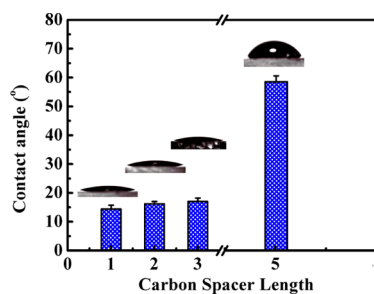


Figure 4. Contact angles of water drops on surfaces grafted with poly(*N*-hydroxyalkyl acrylamide) brushes of various carbon spacer lengths.

polyHMAA, polyHEAA, polyHPAA, and polyHPenAA brushes. All surfaces displayed hydrophilic character similar to other protein-resistant surfaces, but surface wettability was changed as the CSL. With the shorter spacer groups of methylene, ethylene, or propylidene, polyHMAA, polyHEAA, and polyHPAA brushes exhibited small contact angles of 14–18°. As the CSL increased to pentylidene, contact angle increased to 52° for polyHPenAA brush. Methylene as a basic building unit is a typical hydrophobic group, and thus increase in CSL (i.e., increase in the number of methylene groups) leads to an increase in a hydrophobicity/hydrophilicity ratio of the polymers and thus the relatively poor wettability.

Four polymer brushes with similar film thickness of 32–35 nm were tested and compared for surface resistance to protein adsorption from three single-protein solutions. It can be seen in Figure 5a that all poly(*N*-hydroxyalkyl acrylamide)-based brushes with various CSLs exhibited <1 ng/cm² protein adsorption from three different protein solutions. In particular, polyHMAA, polyHEAA, and polyHPAA brushes showed undetectable protein adsorption (<0.3 ng/cm², the detection limit of the SPR sensor). PolyHPenAA brush exhibited 0.3, 0.15, and 0.9 ng/cm² for lysozyme, BSA, and fibrinogen, respectively, which were slightly higher than the other three polymer brushes but still below a superlow fouling level that is defined as <5 ng/cm². The four polymer brushes were further challenged with undiluted human blood serum and plasma containing ~ 500 different types of proteins. Figure 5b showed that (i) polyHMAA and polyHEAA brushes can still retain almost zero protein adsorption, (ii) polyHPAA showed a very minor protein adsorption of 0.3 ng/cm² from blood serum and 0.9 ng/cm² from blood plasma, and (iii) among four polymer brushes, polyHPenAA brushes induced the highest protein adsorption of 0.7 ng/cm² from blood serum and 2.1 ng/cm² from blood plasma. Although SPR data indicate that four polymer brushes can be identified as superlow fouling materials, it is also clear to note that the introduction of the long CSL in polymer backbones tends to make polymer brushes more hydrophobic and thus induces more protein adsorption. If CSL is too short, surface solvation between hydroxyl and amide groups may not be sufficient and short monomeric segments have relative higher rigidity, so that both effects could reduce the protein-resistant ability of polymers. Therefore, there should exist an optimal CSL for the best antifouling performance. In our cases, because of the detection limit of SPR, we did not observe significant difference in protein-resistance ability between polyHEAA and polyHMAA, but SFG data in Figure 6 indeed showed that polyHEAA had a better water binding ability than polyHMAA, particularly at a dry state. We expect that the polyHEAA could be more protein-resistant than polyHMAA under certain conditions.

Interfacial water at polymer interface plays an important role in antifouling performance.^{1,2} To relate surface hydration characteristics to antifouling performance, we applied SFG to quantify the state of interfacial water at different polymer interfaces under both hydrated and dried conditions. Figure 6a showed that all polyHMAA, polyHEAA, and polyHPAA brushes exhibited similar features and peak locations in SFG spectra. PolyHPenAA brush had relatively higher surface hydrophobicity, leading to a reduced SFG signal intensity as compared with signal intensities from the other three brushes.

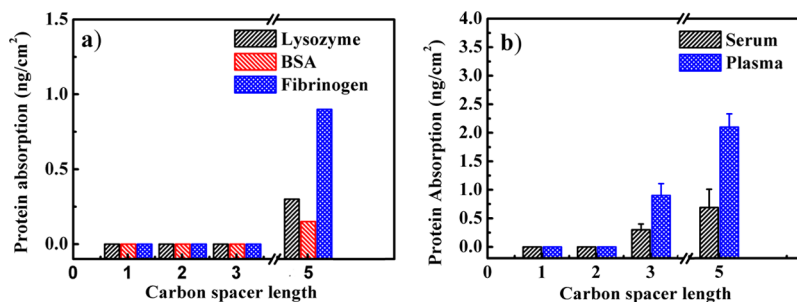


Figure 5. Average protein absorption from (a) three single protein solutions and (b) undiluted human blood plasma and serum on surfaces grafted with poly(*N*-hydroxyalkyl acrylamide)-based brushes of various carbon spacer lengths ($n = 3$).

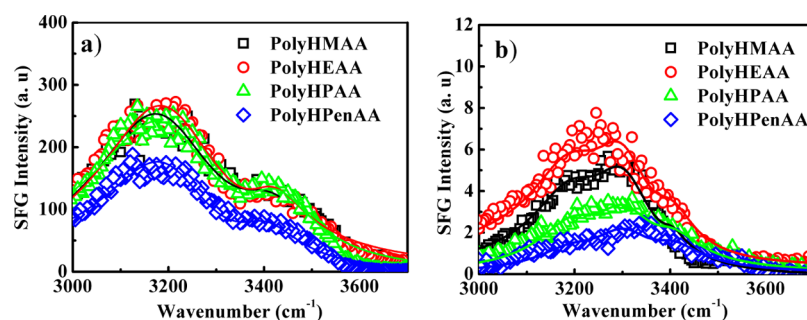


Figure 6. SFG spectra of polyHMAA, polyHEAA, polyHPAA, and polyHPenAA brushes (a) at a wet state and (b) at a dry state.

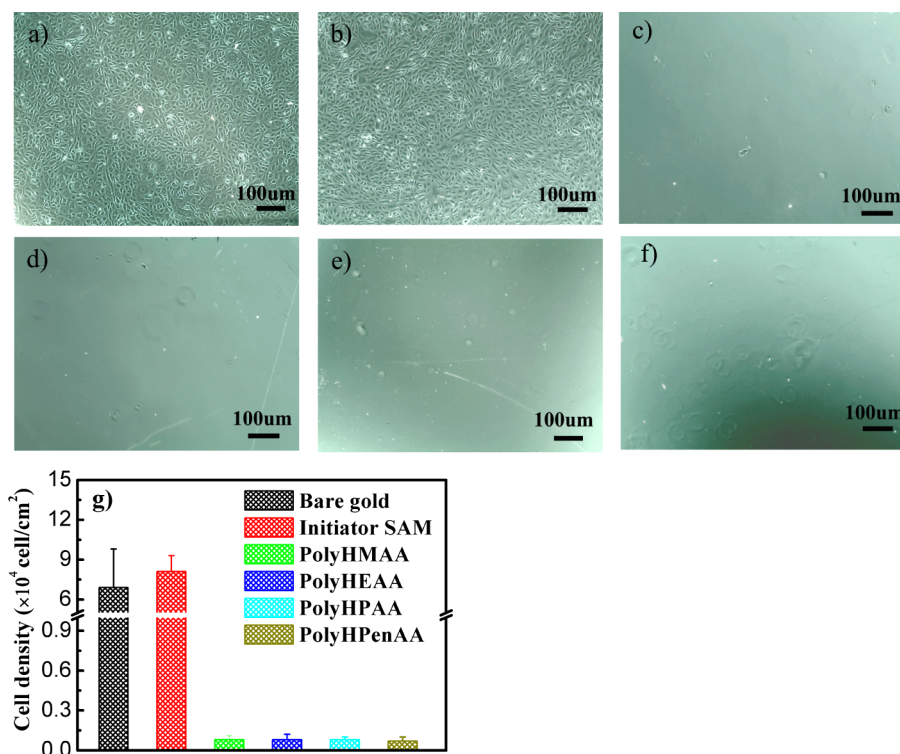


Figure 7. Cell attachment on (a) bare gold, (b) initiator-coated SAM, (c) polyHMAA, (d) polyHEAA, (e) polyHPAA, and (f) polyHPenAA brushes and (g) average densities of BAEC cells adhered on the above polymer brushes ($n = 3$).

All SFG spectra can be fitted with two peaks: a dominant peak at 3200 cm^{-1} that is attributed to water molecules strongly bonded and a weak shoulder-like peak at 3400 cm^{-1} that is attributed to water molecules relatively weakly hydrogen bonded.³⁸ To test the ability of polymer brushes to retain interfacial water, we again performed dry SFG experiments on all polymer brushes. The wet polymer brushes were carefully dried with nitrogen to remove water molecules on surfaces, followed by immediate SFG measurement. Figure 6b clearly shows the CSL dependence of polymers on SFG spectra. Upon drying, all SFG intensities were significantly reduced to a level of below 10 au due to the loss of interfacial water. The overall SFG intensities of the “dry” polymer brushes were in the following order: polyHEAA > polyHMAA > polyHPAA > polyHPenAA. It appears that polyHEAA and polyHMAA (shorter CSL = 1 and 2) exhibit the stronger ability to attract and hold interface water in both wet and dry brushes than polyHPAA and polyHPenAA (longer CSL = 3 and 5). The SFG results demonstrate that different CSLs greatly affect the binding affinity of interfacial water with polymer brushes,

providing further evidence of a correlation between surface hydration and antifouling performance. Jiang and coworkers¹⁸ have studied the influence of CSL in zwitterionic poly-(carboxybetaine) (polyCBAA) on its antifouling performance. They also found that polyCBAA-2 exhibited the better protein-resistance ability than polyCBAA-1, polyCBAA-3, and polyCBAA-5. However, we should note that the CSLs in zwitterionic polyCBAA and hydrophilic polyHAAA act differently to affect polymer hydration. The CSLs in zwitterionic polyCBAA affect the extent of protonation/deprotonation states of polyCBAA as a function of ionic strengths and pH that in turn affect their water binding ability via ionic solvation, while the CSLs in our hydrophilic polyHAAA affect their water binding ability via hydrogen bonds.

3.3. Cell Adhesion on Polymer Brushes of Varied CSLs. It is generally accepted that initial nonspecific protein adsorption on a surface can lead to subsequent cell adhesion and biofilm formation, but a surface that resists protein adsorption does not necessarily imply that this surface can resist cell adhesion as well. Herein, we further studied the adhesion of

endothelial cells onto the four polymer brushes. BAECs were cultured with different polymer brushes at 37 °C for 7 days. Bare gold chips and the chips with initiators were also studied for control. As expected, the cells quickly attached, proliferated, and formed confluent monolayers on bare gold surface and initiator-coated SAM (Figure 7a,b). However, there were almost no cells adhered to the four poly(*N*-hydroxyalkyl acrylamide)-based brushes for 1 week (Figure 7c–f). Figure 7g provides the quantitative data of the adherent cell density on these surfaces (i.e., number per area). The adherent cell densities on bare gold and initiator SAM were $(6.9 \text{ and } 8.1) \times 10^4 \text{ cell/cm}^2$, respectively. Upon grafting poly(*N*-hydroxyalkyl acrylamide)-based brushes, cell densities were dramatically reduced to $(\sim 0.08 \text{ to } 0.1) \times 10^4 \text{ cell/cm}^2$, further confirming that these polymer brushes are also highly resistant to cell adhesion. In addition, we have also tested the cytotoxicity of these polymers themselves. We cultured cells on a gold surface, which was partially coated by polyHAAA. A microscopy image of cell attachment clearly showed very different cell attachment behaviors on the polymer-treated and untreated areas of the surface. The cells were well proliferated and adhered on the untreated area, while they were almost completely absent on the treated area (Figure S2 in the Supporting Information). This confirms that the polymer brushes were not toxic to the cells.

3.4. Molecular Simulations of Polymer Hydration. To provide a better understanding of the CSL effect on the hydration of poly(*N*-hydroxyalkyl acrylamide) monomers and oligomers at atomic level, we investigated the structural, dynamic, and energetic properties of hydrated water molecules associated with four different polymers of various CSLs using quantum mechanics and MD simulations. Figure 8 shows the water–polymer complex structures and their electron density distributions at both HOMO (highest occupied molecular orbital energy) and LUMO (lowest occupied molecular orbital energy) states. It can be seen that both occupied and unoccupied molecular orbitals of four polymers were sited

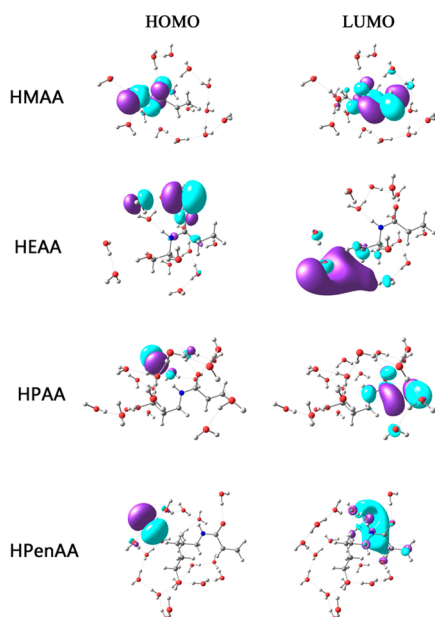


Figure 8. Complex structures of water with HMAA, HEAA, HPAA, and HPenAA as well as their electron density distributions at both HOMO and LUMO states.

with the surrounding water molecules in both HOMO and LUMO states, indicating that the electron transfer between the polymer and surrounding water occurs for all systems. This similarity suggests that variation of the CSL does not significantly affect the interplay between the polymer and the surrounding water. This is probably because, regardless of CSLs, the presence of both amide and hydroxyl groups in polymer enables the formation of hydrogen bonds with the water somehow, generating the more stable polymer–water complex structures. Different polymers displayed different spatial distribution of water molecules around them. In particular, the HEAA–water system presented obvious charge transfer of water from the non-hydrogen-bonding region to the hydrogen-bonding area upon excitation. We may expect that water molecules would form the more potent hydrogen bonds with HEAA and consequently construct a stronger solvent shell around HEAA. Meanwhile, the presence of a long methylene chain (CSL = 5) also resulted in the apparent loss of the charge in the hydrogen-bonding sites upon excitation, and this may lead to the easy breakage of surface hydrogen bonds and a weaker antifouling property. When the $\text{CSL} < 3$, water molecules formed hydrogen bonds bridging between amide and hydroxyl groups to enhance polymer hydration ability. Similar bridging hydrogen-bonded structures between two adjacent oxygen atoms for the PEG chain were also observed, where PEG contains ethylene group ($\text{CSL} = 2$).^{39–42} However, when the $\text{CSL} \geq 3$, bridging hydrogen-bonded structures no longer existed.

We also performed a series of 20 ns MD simulations to evaluate the hydration of four polymers on a longer time scale and a larger length scale. Figure 9 shows the radial distribution

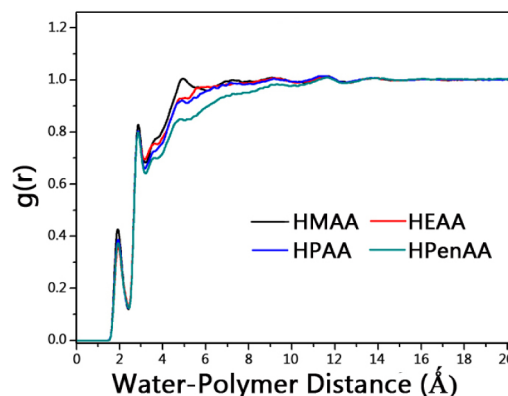


Figure 9. Radial distribution function (RDF) of water molecules around 16-mers of HMAA, HEAA, HPAA, and HPenAA.

functions (RDFs, or $g(r)$) of water molecules around all hydrogen-binding atoms of polymer 16-mers. It can be seen that three RDF profiles of polyHMAA, polyHEAA, and polyHPAA were almost identical to each other, all of which displayed the first hydration peaks at the position of $\sim 2.1 \text{ Å}$ and the second hydration peaks at $\sim 3.2 \text{ Å}$. The nearly identical peak positions for these three polymers suggest that water molecules form similar hydration shells around these polymers. $g(r)$ of polyHPenAA was slightly lower than that of other three polymers, reflecting the reduced hydration shells of polyHPenAA. Our experimental data showed that all of polymer brushes ($\text{CSL} = 1\text{--}5$) exhibited highly surface resistance to nonspecific protein adsorption from undiluted blood plasma and serum at an ultralow fouling level (i.e., $<5 \text{ ng/cm}^2$ of

adsorbed proteins). Only very minor difference of $<2.5 \text{ ng/cm}^2$ in protein adsorption was observed between polymer brushes of CSL = 1–3 and polymer brushes of CSL = 5. Consistently, $g(r)$ profiles also reflect such minor difference among CSL = 1–5 polymers, while all of them are highly hydrated, as evidenced by the two identical peaks occurring at early positions of 2.1 and 3.2 Å. Although $g(r)$ differences appear to be very subtle, considering the fact that only 16-mers were used in MD simulations, the hydration effect would be enlarged as polymer molecular weight increases to experimental scale. We should add cautious remarks that although our simulations of polymers in solution do not truly reflect the surface hydration of polymer brushes, they provide a structure–property relationship of polymers in solution, which can be used as a simple tool to quickly evaluate the antifouling potential of a given materials. Alternatively, simulating polymer brushes on the surface usually requires advanced sampling techniques (e.g., replica-exchange MD and configuration-biased Monte Carlo), molecular models (e.g., coarse-grained polymer model), and, more importantly, the knowledge of surface packing density of polymer chains obtained from experiments or determined by simulations to capture the conformational and dynamic behavior of polymers and interfacial water on the longer time scale and the larger length scale.

4. CONCLUSIONS

In this work, we synthesized a series of poly(*N*-hydroxyalkyl acrylamide)-based materials of polyHMAA, polyHEAA, polyHPAA, and polyHPenAA with different CSLs, with particular attention to the evaluation of the effect of CSLs on the hydration and antifouling ability of these polymer systems using both experimental and computational approaches. SPR data showed that all polymer brushes exhibit excellent antifouling ability to repel proteins from undiluted human blood serum/plasma to a superlow fouling level ($<5 \text{ ng/cm}^2$). These brushes were also highly resistant to cell attachment up to 7 days. SFG results further revealed that polyHMAA and polyHEAA with the shorter CSLs can retain and hold interfacial water more effectively than polyHPMA and polyHPenAA in both wet and dry states, consistent with the enhanced antifouling ability of polyHMAA and polyHEAA. Molecular simulations confirmed that while all polymers presented a strong hydration overall, four polymers displayed different hydration characteristics. Collective data suggest that the antifouling performance of polymers is positively correlated with their hydration properties. The increase of CSLs between amide and hydroxyl groups decreased surface wettability, reduced binding affinity of interfacial water with polymer brushes, and slightly deteriorated antifouling performance. These results provide important structure-based design principles that combination of hydroxyl and amide groups with optimal CSLs can help to obtain effective antifouling materials.

■ ASSOCIATED CONTENT

Supporting Information

Estimation of graft density of polymer brushes. Cytotoxicity of polymers themselves. Retention time of different polymer prepared by adding free initiator. Surface characterization of polymer brushes. Cell adhesion on the chip partially coated by polyHAAA brushes. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The work was funded by National Natural Science Foundation of China (NSFC no. 51273178). J.Y. is grateful for the support from Zhaohui program of Zhejiang University of Technology. Z.C. acknowledges the support from Office of Naval Research (N00014-12-1-0452). J.Z. acknowledges the support from the National Science Foundation grants (CAREER Award CBET-0952624 and CBET-1158447).

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